Robust Few-Shot Aerial Image Object Detection via Unbiased Proposals Filtration

Lingjun Li, Xiwen Yao, *Member, IEEE*, Xue Wang, Dongpao Hong, Gong Cheng, *Member, IEEE*, and Junwei Han

***Abstract*— Few-shot aerial image object detection aims to rapidly detect object instances of novel category in aerial images by using few labeled samples. However, due to the complex background of aerial images, few labeled samples of novel categories, and the model trained with the few-shot learning paradigm is biased toward the base categories, it greatly increases the difficulty of identifying foreground objects of novel categories. In addition to this, tiny object detection is always a hot potato in aerial image object detection, and it is even more difficult for few-shot object detection. To this end, we propose a few-shot aerial image object detection with confidence-iou collaborative proposal filtration and tiny object constraint loss (FsCIT). Specifically, we first introduce a new confidence-iou collaborative proposal filtration scheme to region proposal network (RPN), which combines the unbiased intersection over union (IoU) scores between the two bounding boxes with foreground–background confidence scores to filter redundant region proposals and rescue more foreground proposals for the novel categories in RPN. Then, we design a new tiny object loss constraint term to attempt at overcoming the challenge of tiny object detection in few-shot aerial image object detection. This term considers the central point distance, the size of ground-truth bounding boxes, and the distances between the four edges of the ground-truth bounding box and the predicted bounding box. Experiments on object detection in optical remote sensing images (DIOR), tiny object detection for aerial images (AI-TOD), and high-resolution remote sensing detection (HRRSD) datasets show that FsCIT is effective and can improve the performance of few-shot aerial image object detection.**

***Index Terms*— Aerial images, few-shot object detection (FSOD), tiny object.**

I. Introduction

FEW-SHOT object detection (FSOD) is a research direction attracting much attention in the aerial image field [1], [2], [3], [4], [5], [6], which can effectively alleviate the deep learning problems of high dependence on the amount of annotation data and ignoring unseen categories (i.e., novel categories). FSOD is aimed at achieving the classification and location of object instances of novel category by using a small amount of annotation information [7], [8], [9], [10], [11], [12], [13]. At present, there are two mainstream FSOD learning schemes. One is to use meta-learning to learn meta-knowledge from different meta-tasks and then generalize meta-knowledge to new categories [14], [15], [16], [17], [18], [19], [20]. However, this learning scheme is complicated, time-consuming, and requires elaborate data organization. The other is the finetune-based learning scheme, which is mainly divided into two stages [21], [22], [23], [24], [25], [26]. The first stage is base training, which mainly uses the base categories to train the model, thereinto each class has sufficient labeled samples. The second stage is few-shot fine-tuning, using the data of base and novel categories to train the model, and each class only has a small number of labeled samples. After these two stages of training, the model can identify and locate the object instances of the novel categories. The finetune-based learning scheme is simple and effective and can compare favorably with the meta-learning scheme. In this article, we mainly utilize the finetune-based learning scheme to improve the performance of few-shot aerial image object detection. Meanwhile, we take faster region-based convolution neural network (R-CNN) [27] as the base detector, which is commonly used in most FSOD approaches, no matter in the meta-learning scheme or the finetune-based learning scheme.

In faster R-CNN, the proposals generated in the region proposal network (RPN) are filtered by the nonmaximum suppression (NMS) [28] according to the foreground–background confidence scores output by the RPN. However, the background of aerial image is complicated and the distribution of object instances is scattered, and it is difficult to distinguish foreground object in RPN only by relying on foreground–background scores. In addition, since there are few labeled samples of novel categories and the model trained by the few-shot learning paradigm is biased to base categories, the difficulty of identifying the foreground objects of the novel categories is greatly increased. As shown in Fig. 1(a), the number of the foreground proposals generated by the RPN belong to the novel categories is accounted for very little, which will lead to a decrease in the number of the object features of novel classes extracted in the subsequent the R-CNN head, greatly affecting the detection accuracy of novel categories.

In addition, it is difficult to detect tiny objects in aerial images, as shown in Fig. 1(c), which is even more tricky

Manuscript received 17 November 2022; revised 17 February 2023 and 12 June 2023; accepted 23 July 2023. Date of publication 31 July 2023; date of current version 14 August 2023. This work was supported in part by the National Science Foundation of China under Grant 62071388, Grant 62136007, Grant 62036005, and Grant U20B2068; in part by the Key Research and Development Program of Shaanxi Province under Grant 2023-YBGY-224; and in part by the Innovation Foundation for Doctor Dissertation of Northwestern Polytechnical University under Grant CX2021022. *(Corresponding author: Xiwen Yao.)*

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Digital Object Identifier 10.1109/TGRS.2023.3300071

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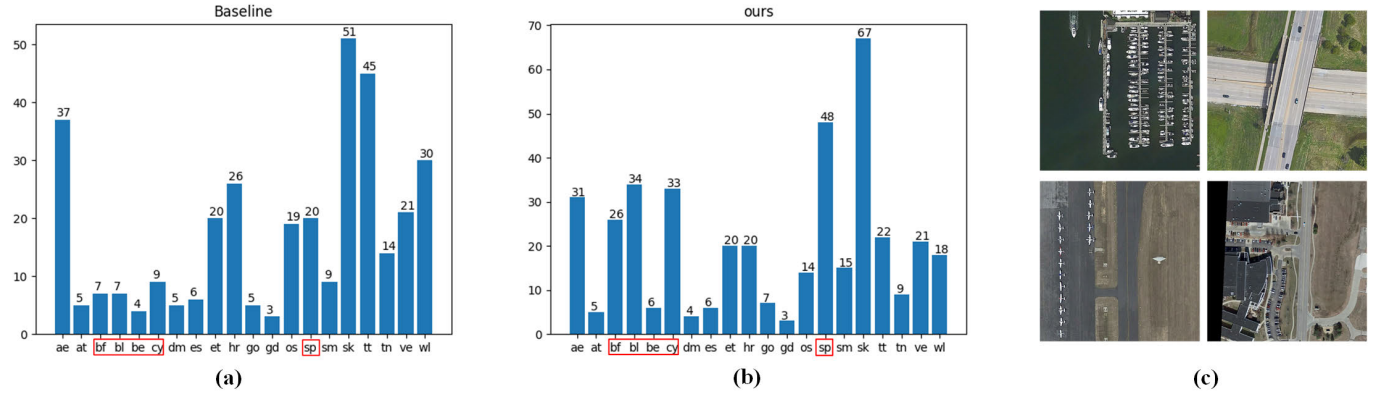


Fig. 1. (a) Baseline: the number of the proposals for each category generated by the RPN in the DIOR dataset. We use 60 images to perform the experiment, including all 20 categories in DIOR dataset. To save space, category names are abbreviated: ae-airplane, at-airport, bf-baseball field, bl-basketball court, be-bridge, cy-chimney, dm-dam, es-expressway service area, et-expressway toll station, hr-harbor, go-golf field, gd-ground track field, os-overpass, sp-ship, sm-stadium, sk-storage tank, tt-tennis court, tn-train station, ve-vehicle, and wl-windmill. Among them, bf, bl, be, cy, and sp are novel categories, and the rest are base categories. (b) Ours: the number of the proposals for each category generated by the RPN in the DIOR dataset. This experiment is performed with the same 60 images as in (a). Compared with the baseline, the number of novel categories of proposals increases significantly. (c) Tiny objects in aerial images.

under the few-shot aerial image object detection. Some efforts have been made to improve the performance of tiny object detection to enhance the resolution of tiny object features by enlarging the size of input images or reducing the downsampling rate of CNN. The feature pyramid network (FPN) [29] and its variants [30], [31], [32], [33] are also proposed to improve the robust learning of the model to multiscale objects.

Indeed, these methods improve the performance of tiny object detection to a certain extent but increase the computational burden and the complexity of the model. More recently, some works [34], [35], [36], [37], [38] have been done by designing elaborate loss functions, such as normalized Wasserstein distance (NWD) loss [39], Kullback–Leibler divergence (KLD) loss [40] by using Gaussian distribution, and dot distance (DotD) loss [41] by considering the size of the object, to overcome the challenge of tiny object detection in aerial images. In this article, due to the small number of labeled samples of novel categories in few-shot aerial image object detection, it is not suitable to devise complex model structure. Therefore, we follow such relatively concise loss design manner to overcome the challenge of tiny object detection under the few-shot aerial image object detection.

To address the above issues, we propose an effective method, named few-shot aerial image object detection with confidence-iou collaborative proposal filtration and tiny object constraint loss (FsCIT). First, we introduce a new confidence-iou collaborative proposal filtration scheme in the RPN stage, including base training stage and few-shot fine-tuning stage. Specifically, in addition to filtering part of proposals by relying on the foreground–background confidence scores generated by RPN, we calculate the intersection over union (IoU) scores between the proposals and ground-truth bounding boxes and use IoU scores to participate in the proposal filtration. On the one hand, the foreground objects in aerial image can be accurately identified under the dual effects of foreground–background classification scores and IoU scores. On the other hand, IoU scores are category-agnostic and do not skew toward base categories, thus preventing foreground proposals belonging to novel categories from being filtered out and retaining more object instance features of the novel categories [see Fig. 1(b)]. By using such a proposal filtration scheme, more foreground features can be extracted from the complex background of aerial images, which is conducive to the feature learning and extraction of aerial images for the FSOD model. This new proposal filtration is only used during the model training phase and will be removed during inference. Second, aiming at the difficulty of tiny object detection in aerial images, we design a new tiny object constraint term and added this constraint term to box regression loss function. This loss constraint term takes into account the central point distance, the size of ground-truth bounding boxes, and the distances between the four edges of the ground-truth bounding box and the predicted bounding box to improve the focus on tiny object during the optimization learning of the FSOD model.

The main contributions of our FsCIT method can be concluded as follows.

1) We propose a novel confidence-iou collaborative proposal filtration scheme to retain more foreground proposals of novel categories and improve the detection accuracy of novel categories in few-shot aerial image object detection.

2) We design a novel tiny object regression constraint loss term to improve the performance of few-shot tiny object detection in aerial images.

3) Our FsCIT can be plugged into existing two-stage FSOD model without changing the model structure. Our method achieves excellent performance on object detection in optical remote sensing images (DIOR), AI-TOD, and high-resolution remote sensing detection (HRRSD) aerial image datasets.

The rest of this article is organized as follows. Section II gives a brief review of FSOD and tiny object detection in aerial images. In Section III, we describe the proposed FsCIT method in detail. Section IV presents the comprehensive experiments on DIOR, AI-TOD, and HRRSD aerial image datasets. Finally, Section V provides the conclusion of this article.

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II. Related Works

In this section, we will review related works in two aspects: FSOD and tiny object detection in aerial images.

A. Few-Shot Object Detection

FSOD has drawn extensive attention in natural image and aerial image fields. Two popular learning schemes, meta-learning and the finetune-based learning, have been widely used to achieve FSOD. LSTD [8] is low-shot transfer detector that integrates SSD [42] and faster R-CNN into a learning framework and designs an elaborate transfer learning scheme for achieving low-shot detection. Meta R-CNN [14] puts forward a predictor-head remodeling network, adds it to faster R-CNN, and applies meta-learning to train the detector. Fan et al. [20] design an attention-RPN and a multirelation module to enable the few-shot detection model to detect novel categories without retraining and fine-tuning. TFA [21] is a two-stage fine-tuning method that has superiority over other previous meta-learning approaches. Few-shot object detection via contrastive proposals encoding (FSCE) [25] and decoupled faster R-CNN (DeFRCN) [26] are based on the improvement of TFA, and both believe that it is not enough to fine-tune only the last layer of detector, and the fine-tuning of other modules should also be considered. Cheng et al. [1] propose a prototype-CNN that contains a prototype learning network and a prototype-guided RPN to improve the performance of FSOD in remote sensing images. In our work, we follow the finetune-based learning scheme and improve the performance of few-shot aerial image object detection without increasing the complexity of model structure.

B. Tiny Object Detection in Aerial Images

There are a large number of tiny objects in aerial images, resulting in limited detection performance. Some works improve tiny object detection by enhancing feature learning. For example, SSD locates and classifies the objects of various sizes by using image feature maps with different resolutions. FPN fuses features of different scales by designing a top–down with lateral connections network structure. Recursive-FPN [43], BiFPN [44], and PANet [30] are all improvements to FPN. In addition, there are some works using generative adversarial network (GAN) [45] to generate superpixel representations of tiny object. Both perceptual GAN [46] and small object detection via multi-task generative adversarial network (SOD-MTGAN) [47] utilize GAN to produce superresolution representation for tiny object, recovering more representation information of tiny object. Differently, scale normalization for image pyramids (SNIP) [48] and SNIP with efficient resampling (SNIPER) [49] are two well-designed training strategies to enhance tiny object detection. Recently, some works have tried to solve the problem of tiny object detection by designing specialized loss functions and object box representation. Wang et al. [50] first propose a new AI-TOD dataset and construct a M-CenterNet to detect tiny object. DotD [41] uses the relationship between the average scale of all objects in a dataset and the central distance of the two bounding boxes to replace the previous IoU loss and realize the detection performance improvement of tiny objects. Wang et al. [39] propose an NWD to calculate the similarity between the two bounding boxes by their Gaussian distributions. Our work follows the idea of designing loss function, because this kind of solution does not need to design complex model structure and training strategy, and does not significantly increase the computational burden of the model.

III. Proposed Method

A. Preliminary

Following previous work setting, we divide the categories in a dataset into base categories $\mathcal{C}_b$ and novel categories $\mathcal{C}_n$. There is no intersection between $\mathcal{C}_b$ and $\mathcal{C}_n$, i.e., $\mathcal{C}_b \cap \mathcal{C}_n = \emptyset$. Let $\mathcal{D}_b$ denotes a subdataset, which contains only the base categories $\mathcal{C}_b$, each of base class has abundant annotated samples. There are only novel categories $\mathcal{C}_n$ in the subdataset $\mathcal{D}_n$ and a small number of annotated data for each novel class.

In this work, we propose a few-shot aerial image object detection with FsCIT. On the one hand, we design a new confidence-iou collaborative proposal filtration scheme to overcome the difficulty of foreground object recognition in few-shot aerial image object detection and rescue more foreground proposals for novel categories $\mathcal{C}_n$; on the other hand, we construct a new tiny object loss constraint term to attempt at overcoming the challenge of tiny object detection in few-shot aerial image object detection. In FsCIT, tiny object constraint loss (TOCL) and confidence-iou collaborative proposal filtration are only applied in the training stage of the model, including base training and few-shot fine-tuning stage, but not in the inference stage, which will not increase the time cost of model inference.

B. Confidence-IoU Collaborative Proposal Filtration

We propose a new confidence-iou collaborative proposal filtration scheme of RPN. Generally, in RPN, many of the proposals are filtered out depending on the foreground–background scores output by the RPN, and then, NMS is used to choose the final proposals from the remaining. These final proposals are fed into the subsequent R-CNN head. In our confidence-iou collaborative proposal filtration scheme, besides relying on the foreground–background scores to filter the redundant proposals, we also use the IoU scores between proposals and ground-truth bounding boxes to select proposals. Mathematically, assuming that the number of proposals to be preselected according to the foreground–background scores and IoU scores before NMS is $\mathbf{N}_{\text{pre}}$, where the number of proposals determined by the foreground–background scores is n^{s}_{pre}, and the number of proposals made by the IoU scores is $n^{\text{ious}}_{\text{pre}}$, $\mathbf{N}_{\text{pre}} = n^{s}_{\text{pre}} + n^{\text{ious}}_{\text{pre}}$. Ranking the foreground–background scores $\text{scores}^{\text{fb}}$ in descending order

$$\text{scores}_{\text{index}} = \text{Sort}\left(\text{scores}^{\text{fb}}\right). \tag{1}$$

The first n^{s}_{pre} proposals are selected from the initial set of proposals $\mathbf{P}_{\text{initial}}$ according to the sorted indexes $\text{scores}_{\text{index}}$, denoted as $\mathbf{P}^{s}_{\text{pre}}$. Then, for the rest of the proposals $\mathbf{P}_{\text{remaining}}$, we calculate the IoU scores $\text{scores}^{\text{IoU}}$ between $\mathbf{P}_{\text{remaining}}$ and matched ground-truth bounding boxes $\mathbf{GT}_{\text{remaining}}$ and sort IoU scores $\text{scores}^{\text{IoU}}$

$$\text{IoU}_{\text{index}} = Sort\left(\text{scores}^{\text{IoU}}\right). \tag{2}$$

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Algorithm 1 Confidence-IoU Collaborative Proposal Filtration

Input: $\mathbf{P}_{initial}$, $\mathbf{GT}^{matched}$, $scores^{fb}$, n^{s}_{pre}, n^{ious}_{pre}, n^{s}_{post}, n^{ious}_{post}
Output: $\mathbf{P}_{final}$

```
for image_index = 1, 2, ..., N do
    Sort scores^fb using Eq. (1)
    Select P^s_pre from P_initial by scores_index and n^s_pre
    Calculate scores^IoU between P_remaining and GT_remaining
    Sort scores^IoU using Eq. (2)
    Select P^IoU_pre from P_remaining by IoU_index and n^ious_pre
    P^s_post ← NMS(P^s_pre, n^s_post)
    P^IoU_post ← Sort(P^IoU_pre, n^ious_post)
    P_final ← append(P^s_post ∪ P^IoU_post)
end for
```

We use these sorted indexes $\mathrm{IoU}_{\mathrm{index}}$ to select the first $n^{\mathrm{ious}}_{\mathrm{pre}}$ proposals from $\mathbf{P}_{\mathrm{remaining}}$, denoted as $\mathbf{P}^{\mathrm{IoU}}_{\mathrm{pre}}$. Then, assuming that the number of proposals to be selected according to the foreground–background scores and IoU scores after NMS is $\mathbf{N}_{\mathrm{post}}$. The proposals $\mathbf{P}^{s}_{\mathrm{pre}}$ are processed by NMS, and we get n^{s}_{post} of them, denoted as $\mathbf{P}^{s}_{\mathrm{post}}$. Analogously, we reorder the IoU scores of the proposals $\mathbf{P}^{\mathrm{IoU}}_{\mathrm{pre}}$ and select the first n^{ious}_{post} proposals from $\mathbf{P}^{\mathrm{IoU}}_{\mathrm{pre}}$, denoted as $\mathbf{P}^{\mathrm{IoU}}_{\mathrm{post}}$. Finally, $\mathbf{P}^{s}_{\mathrm{post}}$ and $\mathbf{P}^{IoU}_{\mathrm{post}}$ are integrated together as the final set of proposals $\mathbf{P}_{\mathrm{final}}$ and sent to the subsequent R-CNN head. We provide the learning process of confidence-iou collaborative proposal filtration in Algorithm 1. Our scheme can be used in the whole training process of model and can improve the performance more than using the few-shot fine-tuning stage alone (more details in Section IV-C).

C. Tiny Object Constraint Loss

Tiny object detection is always a hard nut to crack in aerial image object detection, and it is even more difficult for FSOD. In order to improve the accuracy of tiny object, we add an additional tiny object constraint item to the existing IoU loss functions. This tiny object constraint item takes into account the central point distance, the size of ground-truth bounding boxes, and the distances between the four edges of the ground-truth bounding box and the predicted bounding box to guide the gradient of the loss function to optimize in a direction more conducive to tiny objects. Mathematically, let $\mathbf{B} = \{x_b, y_b, w_b, h_b\}$ be a predicted bounding box and $\mathbf{G} = \{x_g, y_g, w_g, h_g\}$ as a ground-truth box. $\{x_b, y_b\}$ (or $\{x_g, y_g\}$) is the central point of the box $\mathbf{B}$ (or $\mathbf{G}$), and $\{w_b, h_b\}$ (or $\{w_g, h_g\}$) is the width and height of the box $\mathbf{B}$ (or $\mathbf{G}$). The IoU loss is written as

$$L_{\mathrm{IoU}} = 1 - \mathrm{IoU}, \mathrm{IoU} = \frac{\mathbf{B} \cap \mathbf{G}}{\mathbf{B} \cup \mathbf{G}}. \tag{3}$$

On the basis of IoU loss, we add a tiny object constraint item L_{tiny}, namely

$$L_{\mathrm{IoU}} = 1 - \mathrm{IoU} + L_{\mathrm{tiny}}. \tag{4}$$

With (4), we can generalize to other IoU-based losses, such as GIoU [51], DIoU [52], CIoU [53], and so on. L_{tiny} is defined as

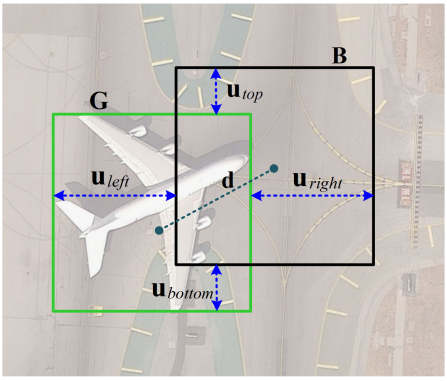


Fig. 2. Illustration of some distance variables in the tiny object constraint item.

$$L_{\mathrm{tiny}} = \alpha \cdot \phi(\mathbf{S}_g) \cdot \left(\frac{\mathbf{d}^2}{\mathbf{S}_g} + \frac{\mathbf{u}_{\mathrm{top}} + \mathbf{u}_{\mathrm{bottom}}}{\mu \cdot h_g} + \frac{\mathbf{u}_{\mathrm{left}} + \mathbf{u}_{\mathrm{right}}}{\mu \cdot w_g} \right) \tag{5}$$

where α and μ are the hyperparameters, and $\mathbf{S}_g$ is the area of the ground-truth box $\mathbf{G}$. $\phi(\mathbf{S}_g)$ is written as

$$\phi(\mathbf{S}_g) = e^{-\frac{\mathbf{S}_g - \min(\mathbf{S}_g)}{\max(\mathbf{S}_g) - \min(\mathbf{S}_g)}}. \tag{6}$$

As shown in Fig. 2, $\mathbf{d}$ is the distance between the central points of $\mathbf{B}$ and $\mathbf{G}$

$$\mathbf{d} = \sqrt{(x_g - x_b)^2 + (y_g - y_b.)^2} \tag{7}$$

$\mathbf{u}_{\mathrm{top}}$, $\mathbf{u}_{\mathrm{bottom}}$, $\mathbf{u}_{\mathrm{left}}$, and $\mathbf{u}_{\mathrm{right}}$ are the distances between the four edges of $\mathbf{B}$ and $\mathbf{G}$

$$\begin{aligned} \mathbf{u}_{\mathrm{top}} &= \left| y_g - \frac{h_g}{2} - y_b + \frac{h_b}{2} \right|; \mathbf{u}_{\mathrm{bottom}} = \left| y_g + \frac{h_g}{2} - y_b - \frac{h_b}{2} \right| \\ \mathbf{u}_{\mathrm{left}} &= \left| x_g - \frac{w_g}{2} - x_b + \frac{w_b}{2} \right|; \mathbf{u}_{\mathrm{right}} = \left| x_g + \frac{w_g}{2} - x_b - \frac{w_b}{2} \right|. \end{aligned} \tag{8}$$

To explain how the size of tiny object affects the gradient optimization direction, we analyze our tiny object constraint item L_{tiny} in terms of gradient. For $\mathbf{d}$, we have

$$\frac{\partial \left(\frac{\mathbf{d}^2}{\mathbf{S}_g} \right)}{\partial \mathbf{d}} = \frac{2\mathbf{d}}{\mathbf{S}_g}. \tag{9}$$

The weight $1/\mathbf{S}_g$ enables the model to be dynamically optimized according to the object scale. When the object scale $\mathbf{S}_g$ is small, the weight $1/\mathbf{S}_g$ becomes large, which means that the model pays more attention to the optimization of the tiny object. For $\mathbf{u}_{\mathrm{top}}$, $\mathbf{u}_{\mathrm{bottom}}$, $\mathbf{u}_{\mathrm{left}}$, and $\mathbf{u}_{\mathrm{right}}$, we have

$$\frac{\partial \left(\frac{\mathbf{u}_{\mathrm{top}} + \mathbf{u}_{\mathrm{bottom}}}{\mu \cdot h_g} + \frac{\mathbf{u}_{\mathrm{left}} + \mathbf{u}_{right}}{\mu \cdot w_g} \right)}{\partial (\mathbf{u}_{\mathrm{top}}, \mathbf{u}_{\mathrm{bottom}}, \mathbf{u}_{left}, \mathbf{u}_{right})} = \left(\frac{1}{\mu \cdot h_g}, \frac{1}{\mu \cdot w_g} \right). \tag{10}$$

Similarly, when the size of object is small, the weights $1w_g$ and $1/h_g$ become large, which can promote better optimization of the tiny object. Our TOCL function can also be applied to general object detection method for natural images. However, our loss function is more suitable for aerial images, because the size difference between objects in aerial images is larger. Tiny objects can be as small as 2–8 pixels. Compared with

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natural images, such tiny object size can play a more obvious role in our loss function gradient optimization.

In addition, $\phi(\mathbf{S}_g)$ is added so as not to degrade the accuracy of relatively large objects. Because the smaller the object, the larger the parameter $\phi(\mathbf{S}_g)$, the larger the object, the smaller the parameter $\phi(\mathbf{S}_g)$. Because the range of $(1-\text{IoU})$ is $[0, 1]$, for consistency, L_{tiny} is rewritten as

$$L_{\text{tiny}} = \alpha \cdot \phi(\mathbf{S}_g) \cdot \left(2 - e^{-\frac{\mathbf{d}^2}{\mathbf{S}_g}} - e^{-\left(\frac{\mathbf{u}_{\text{top}}+\mathbf{u}_{\text{bottom}}}{\mu \cdot h_g} + \frac{\mathbf{u}_{\text{left}}+\mathbf{u}_{\text{right}}}{\mu \cdot w_g}\right)}\right). \tag{11}$$

IV. Experiments

In this section, we first provide extensive ablation studies of our FsCIT. Then we add our FsCIT on some FSOD methods to validate the effectiveness of our FsCIT.

A. Datasets

DIOR: It is a large-scale benchmark dataset for object detection in optical remote sensing images. DIOR [54] contains 23 463 images and 190 288 object instances, covering 20 categories, including airplane, airport, baseball field, basketball court, bridge, chimney, dam, expressway service area, expressway toll station, harbor, golf course, ground track field, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and windmill. In our experiments, these 20 DIOR categories are randomly divided into 15 base classes and five novel classes. The experimental comparisons are mainly conducted on two different category split sets, i.e., Novel split 1 and Novel split 2, as well as the other datasets introduced later. In Novel split 1, the five novel classes include: baseball field, basketball court, bridge, chimney, ship. The rest are base classes. In Novel split 2, airplane, airport, expressway toll station, harbor, and ground track field are five novel classes. For M-shot of each novel class, we set $M = 10$ and 20, as well as for the datasets introduced later.

AI-TOD: AI-TOD [50] has 28 036 aerial images, 700 621 object instances, and eight categories, including airplane, bridge, storage-tank, ship, swimming-pool, vehicle, person, and windmill. AI-TOD is customized for evaluating the performance of detectors on tiny object detection, so the largest size of objects in this dataset is smaller than 64 pixels, and more than half of the objects are less than 16 pixels. In our experiments, the overall eight categories are randomly split into five base classes and three novel classes. In Novel split 1, the three novel classes include: airplane, swimming pool, and windmill. In Novel split 2, bridge, swimming pool, and storage tank are three novel classes.

HRRSD: HRRSD [55] contains 21 761 images, 55 740 object instances, and 13 remote sensing image categories, including ship, bridge, ground track field, storage tank, basketball court, tennis court, airplane, baseball diamond, harbor, vehicle, crossroad, T junction, and parking lot. In our experiment, we randomly select five categories from all categories as novel categories and regard the remaining as base categories. In Novel split 1, the five novel classes include: ground track field, basketball court, airplane, vehicle, and parking lot. In Novel split 2, ship, storage tank, baseball diamond, harbor, and crossroad are five novel classes.

B. Implementation Details

We perform all the comparison and ablation experiments with a single NVIDIA GeForce GTX 1080 Ti GPU. ResNet-101 pretrained on ImageNet is used as the backbone. SGD is used to optimize the model. We set learning rate as 0.002 in the base training stage, 0.001 in the few-shot fine-tuning stage, and $M = 10$ and 20. In our experiments, we mainly use the GIoU loss function and then add the tiny object constraint item we designed on the basis of this function. We use average precision (AP) metric to evaluate the detection performance on DIOR, AI-TOD, and HRRSD. AP50 (or AP75) indicates the AP measurement value when the IoU threshold is greater than 0.5 (or 0.75). APvt represents the AP measurement value when the size of very tiny objects is 2–8 and APt represents the AP measurement value when the size of tiny objects is 8–16. APs represent the AP measurement value when the size of small objects is 16–32. APm represents the AP measurement value when the size of medium objects is 32–96. APl represents the AP measurement value when the size of large objects is larger than 96.

C. Ablation Analysis

In this section, we use DeFRCN as a baseline and use a GIoU loss for box regression deltas to verify the effectiveness of confidence-iou collaborative proposal filtration (CICPF) and TOCL on DIOR and AI-TOD datasets, respectively. Note that we mainly take Novel split 1 and set the sample number of each novel class to 10.

Ablation on the number of proposals in CICPF: There are two hyperparameters k and t in CICPF, which are used to determine the values of n^s_{pre}, $n^{\text{ious}}_{\text{pre}}$, n^s_{post}, and $n^{\text{ious}}_{\text{post}}$. $\mathbf{N}_{\text{pre}} = n^s_{\text{pre}} + n^{\text{ious}}_{\text{pre}}$ is the number of proposals to be preselected according to the foreground–background scores and IoU scores before NMS. We set $n^s_{\text{pre}} = \lceil \mathbf{N}_{\text{pre}}/k \rceil$. Similarly, $\mathbf{N}_{\text{post}} = n^s_{\text{post}} + n^{\text{ious}}_{\text{post}}$ denotes the number of proposals to be selected according to the foreground–background scores and IoU scores after NMS. We set $n^s_{\text{post}} = \lceil \mathbf{N}_{\text{post}}/t \rceil$. We first study the influence of the hyperparameter k with different values in the case of $t = 4$. As can be observed from Fig. 3, $k = 6$ achieves the best performance boost. Then, we fix $k = 6$ and study the influence of the hyperparameter t with different values. From Table I, we find that the model achieves better performance when $t = 4$. Note that the above experiments are for CICPF only without the participation of TOCL.

Ablation on the hyperparameters of TOCL: In this ablation experiment, two hyperparameters α and μ need to be studied. From Fig. 4, we can see that, with fixed $\mu = 2$, $\alpha = 3$ is an option that can bring better performance improvements. Then, $\alpha = 3$ is fixed, and we study the influence of the hyperparameter μ with different values. In Table II, we find that $\mu = 2$ achieves better performance. Note that the above experiments are for TOCL only, without the participation of CICPF. In order to further verify that TOCL is effective for tiny object detection, we have also carried out experimental

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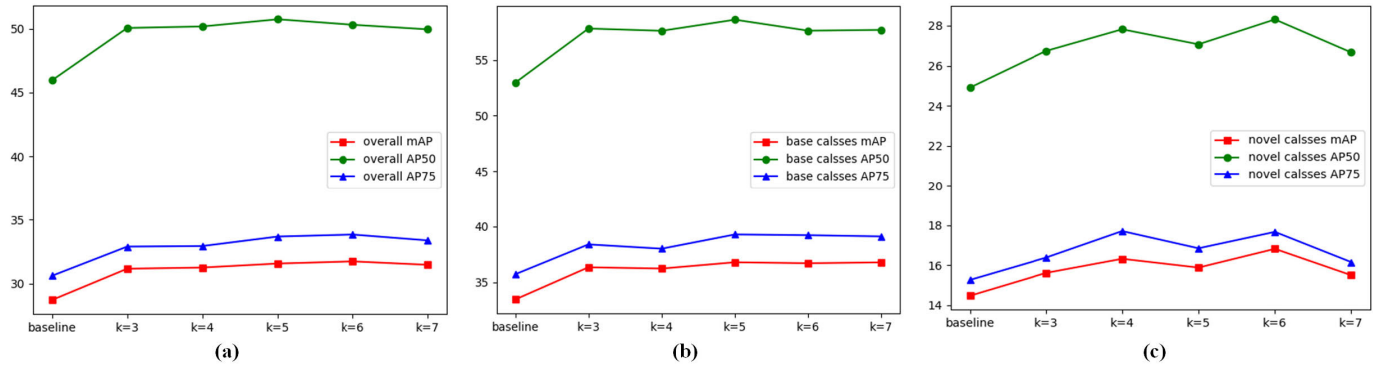


Fig. 3. Accuracies (%) with different hyperparameters k on DIOR. We choose 15 base classes and five novel classes from 20 DIOR categories. Shot numbers are 10. (a) Accuracies on overall 20 classes. (b) Accuracies on base classes. (c) Accuracies on novel classes.

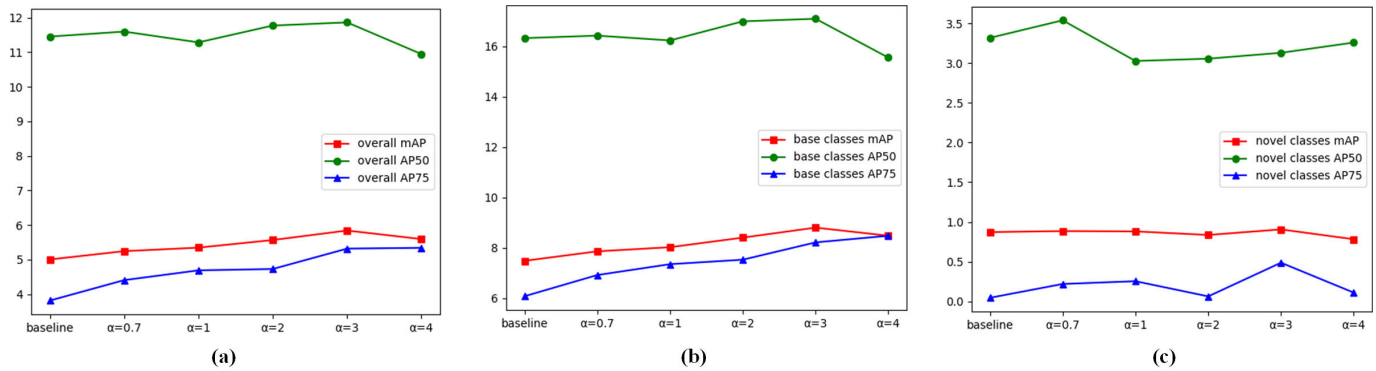


Fig. 4. Accuracies (%) with different hyperparameters α on AI-TOD. We choose five base classes and three novel classes from eight AI-TOD categories. Shot numbers are 10. (a) Accuracies on overall eight classes. (b) Accuracies on base classes. (c) Accuracies on novel classes.

TABLE I

ACCURACIES (%) WITH DIFFERENT HYPERPARAMETERS t ON DIOR. SHOT NUMBERS ARE 10

t	Overall (20) AP	Base (15) AP	Novel (5) AP
Baseline	28.699	33.441	14.471
3	31.336	36.303	16.437
4	**31.727**	36.697	**16.818**
5	31.662	36.796	16.26

TABLE II

ACCURACIES (%) WITH DIFFERENT HYPERPARAMETERS μ ON AI-TOD. SHOT NUMBERS ARE 10

μ	Overall (8) AP	Base (5) AP	Novel (3) AP
Baseline	5.001	7.478	0.872
1	5.56	8.394	0.837
2	**5.838**	8.798	**0.906**
3	5.787	8.866	0.654

comparisons with some other loss functions designed for small object detection tasks on the AI-TOD dataset, and from Table III, the experimental results show that our loss function has better effect for tiny object detection. Note that the above experiments are for TOCL only without the participation of CICPF.

Influence of different components of FsCIT: In this section, we analyze the influence of different components (i.e., CICPF and TOCL) on detection performance. As shown in Table IV, our two components, CICPF and TOCL, achieve superior performance over the baseline. We also carry out experiments to verify that the detection performance obtained by using CICPF in the whole training process is higher than that obtained by using CICPF in the few-shot fine-tuning stage alone.

D. FsCIT Improves Previous Few-Shot Detection Methods

We add our proposed FsCIT on four few-shot two-stage object detection baselines, including TFA [21], generalized few-shot detection (GFSD) [24], FSCE [25], and DeFRCN [26], to validate the effectiveness of our FsCIT.

DIOR: We compare FsCIT with other methods on the DIOR datatset. Table V shows the detection results on the Novel split 1 of DIOR, where the shot number is 10 and 20, respectively. Compared with FSRW [56] and MPSR [23], the other algorithms in Table V use only one GPU for model training and obtain better detection results. FsCIT improves the detection performance of TFA, GFSD, FSCE, and DeFRCN. Taking DeFRCN for an example, when the shot number is 10, DeFRCN with our FsCIT gains about 5.026%, 5.843%, and 2.57% on AP of overall, AP of base, and AP of novel, respectively. For 20-shot, DeFRCN with our FsCIT gains about 5.203%, 5.949%, and 2.967% on AP of overall, AP of base, and AP of novel, respectively. Besides, we also report AP on the Novel split 2 of DIOR. The experimental results shown

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TABLE III

COMPARISON WITH OTHER LOSS FUNCTIONS DESIGNED FOR SMALL OBJECT DETECTION TASKS ON AI-TOD

Method	Overall (8)							Base (5)			Novel (3)		
	AP	AP50	AP75	APvt	APt	APs	APm	AP	AP50	AP75	AP	AP50	AP75
Focal-EIOU[38]	4.986	11.185	3.963	2.258	5.780	5.1	5.9	7.643	16.487	6.292	0.558	2.348	0.082
DIoU[52]	4.998	11.475	3.936	2.820	5.694	4.4	6.2	7.437	16.388	6.212	0.905	3.288	0.142
NWD[39]	4.387	10.640	3.093	1.994	4.807	4.2	7.1	6.485	15.314	4.611	0.890	2.851	**0.563**
GWD[35]	1.989	7.134	0.376	0.667	2.286	2.2	2.4	2.855	10.162	0.572	0.546	2.088	0.050
KLD[40]	5.125	**12.032**	3.845	2.555	5.874	5.0	5.3	7.786	**17.109**	6.100	0.691	**3.569**	0.086
KFIoU[36]	3.386	10.155	1.298	2.876	3.739	3.5	5.2	4.922	14.485	1.966	0.826	2.940	0.185
Smooth L1[37]	4.996	10.747	4.156	2.645	6.048	4.4	6.4	7.657	15.880	6.607	0.559	2.194	0.070
GIoU[51]	5.001	11.450	3.811	2.353	5.608	5.3	5.6	7.478	16.330	6.070	0.872	3.317	0.045
GIoU +TOCL(ours)	**5.838**	11.652	**5.191**	**3.097**	**6.529**	**5.6**	**6.4**	**8.798**	16.735	**8.167**	**0.906**	3.180	0.230

TABLE IV

INFLUENCE OF DIFFERENT COMPONENTS OF FSCIT ON DIOR. "∗" MEANS ONLY USING CICPF IN THE FEW-SHOT FINE-TUNING STAGE

CICPF	TOCL	Overall (20)						Base (15)						Novel (5)					
		AP	AP50	AP75	APs	APm	APl	AP	AP50	AP75	APs	APm	APl	AP	AP50	AP75	APs	APm	APl
		28.70	45.94	30.60	2.47	19.03	46.09	33.44	52.96	35.71	3.39	24.32	53.84	14.47	24.90	15.26	0.24	3.16	22.85
	✓	30.22	46.65	32.64	3.03	20.27	47.88	35.21	53.81	38.09	4.17	25.22	55.88	15.23	25.16	16.27	0.30	5.42	23.90
✓∗		29.06	46.94	30.54	2.53	20.6	46.28	33.34	53.49	34.96	3.47	25.90	53.31	16.21	27.28	17.26	0.26	4.69	25.21
✓		31.73	50.30	33.83	3.85	23.01	49.26	36.70	57.63	39.22	5.40	29.16	56.81	16.82	28.31	17.67	0.13	4.56	26.61
✓	✓	**33.73**	52.19	36.39	4.56	27.17	50.86	**39.28**	60.22	42.40	6.35	34.24	58.70	**17.04**	28.12	18.36	0.26	5.97	27.34

TABLE V

DETECTION RESULTS (%) OF FSOD METHODS ON DIOR. SHOT NUMBERS ARE 10 AND 20 ON NOVEL SPLIT 1

10 Shot	Overall (20)						Base (15)			Novel (5)		
Methods	AP	AP50	AP75	APs	APm	APl	AP	AP50	AP75	AP	AP50	AP75
FSRW[56]	9.342	23.798	5.534	0.066	4.088	20.456	10.546	27.198	6.137	5.732	13.6	3.726
MPSR[23]	6.734	14.985	5.184	0.533	4.4	12.789	6.926	15.907	5.076	6.155	12.22	5.507
TFA[21]	32.292	51.509	34.196	7.172	27.182	47.445	42.142	66.975	44.745	2.74	5.109	2.549
+ FsCIT	**35.766**	**55.387**	**38.163**	**9.863**	**30.227**	**50.545**	**46.415**	**71.256**	**49.805**	**3.818**	**7.779**	**3.236**
GFSD[24]	33.617	55.558	34.964	8.612	27.666	49.42	42.226	68.76	44.395	7.787	15.95	6.672
+ FsCIT	**36.659**	**58.205**	**39.005**	**10.345**	**30.584**	**52.545**	**46.062**	**72.101**	**49.432**	**8.452**	**16.518**	**7.725**
FSCE[25]	27.574	45.065	29.369	2.454	16.913	47.38	32.07	51.779	34.208	14.084	24.923	14.852
+ FsCIT	**30.404**	**47.152**	**33.158**	**2.828**	**20.186**	**50.475**	**35.446**	**54.222**	**38.786**	**15.279**	**25.941**	**16.272**
DeFRCN[26]	28.699	45.943	30.596	2.466	19.028	46.093	33.441	52.956	35.707	14.471	24.902	15.263
+ FsCIT	**33.725**	**52.192**	**36.392**	**4.561**	**27.172**	**50.859**	**39.286**	**60.216**	**42.401**	**17.041**	**28.119**	**18.362**
20 Shot	Overall (20)						Base (15)			Novel (5)		
Methods	AP	AP50	AP75	APs	APm	APl	AP	AP50	AP75	AP	AP50	AP75
FSRW[56]	10.169	25.109	6.431	0.075	4.607	22.678	10.869	26.947	6.884	8.068	19.592	5.073
MPSR[23]	8.887	20.113	6.405	1.048	7.28	16.772	9.682	22.206	6.814	6.504	13.833	5.177
TFA[21]	32.881	52.104	34.931	7.391	27.371	48.254	42.844	67.596	45.675	2.991	5.629	2.701
+ FsCIT	**36.316**	**55.957**	**38.779**	**10.087**	**30.759**	**51.093**	**46.822**	**71.562**	**50.156**	**4.797**	**9.142**	**4.648**
GFSD[24]	33.836	55.733	35.46	8.629	27.854	49.705	42.196	68.68	44.566	8.756	16.891	8.143
+ FsCIT	**37.192**	**58.955**	**39.755**	**10.435**	**31.148**	**53.347**	**46.06**	**72.068**	**49.456**	**10.587**	**19.617**	**10.653**
FSCE[25]	29.752	47.897	32.112	3.072	18.886	49.601	33.66	53.56	36.257	18.027	30.905	19.679
+ FsCIT	**33.318**	**51.185**	**36.536**	**4.338**	**23.559**	**53.646**	**37.629**	**57.473**	**41.003**	**20.386**	**32.32**	**23.136**
DeFRCN[26]	31.483	50.323	33.643	3.424	22.098	48.844	35.668	56.691	37.968	18.927	31.217	20.666
+ FsCIT	**36.686**	**56.515**	**39.654**	**6.317**	**28.943**	**53.48**	**41.617**	**64.032**	**44.652**	**21.894**	**33.965**	**24.66**

in Table VI indicate that FsCIT remains competitive on the Novel split 2 (27.984% DeFRCN versus 33.125% DeFRCN with FsCIT). It can be seen from the experimental results that our FsCIT not only improves the detection performance of novel categories but also improves that of base categories, which also confirms that our CICPF can effectively alleviate the difficulty of foreground recognition in aerial images. We also visualize the detection results in Fig. 5.

AI-TOD: We report the detection results on the Novel split 1 of AI-TOD in Table VII. Due to memory limitations of GPU, we discard the layer P2 in FPN module and start processing features from layer P3. We observe that the detection results on this dataset are obviously not as high as those on DIOR dataset, because most of the object instances on AI-TOD dataset are tiny objects, which are difficult to detect. In this case, by using our FsCIT, the obtained detection performance is improved under various evaluation criteria. Concretely, taking DeFRCN with our FsCIT for an example and ten shot, the AP, respectively, improves by 1.186%, 1.077%, 1.859%, 0.729%, 0.969%, 1%, and 4.3% on AP, AP50, AP75, APvt, Apt, APs, and APm of overall, and 1.579% and 0.532% on AP of base and AP of novel, respectively, which indicates that our method can alleviate the problem of small object detection. Note that even with 20 shot, our FsCIT can still improve the performance of the original algorithms. Similarly, the detection results on the Novel split 2 of AI-TOD are shown in Table VIII. From Table VIII, our FsCIT still remains a high-level performance (4.841% DeFRCN vs. 6.035% DeFRCN with FsCIT). In addition, it is worth noting that FSCE with FsCIT does not improve performance in the case of ten shot of the Novel split 2 and 20 shot of the Novel split 1 in Tables VII and VIII. In view of this situation,

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TABLE VI

DETECTION RESULTS (%) OF FSOD METHODS ON DIOR. SHOT NUMBERS ARE 10 AND 20 ON NOVEL SPLIT 2

Methods	10 shots					20 shots				
	Overall (20)			Base (15)	Novel (5)	Overall (20)			Base (15)	Novel (5)
	AP	AP50	AP75	AP	AP	AP	AP50	AP75	AP	AP
TFA[21]	32.827	50.042	35.499	43.317	1.354	33.652	51.433	36.026	44.152	2.153
+ FsCIT	**35.856**	**54.308**	**38.606**	**46.55**	**3.772**	**36.495**	**55.283**	**39.041**	**47.161**	**4.496**
GFSD[24]	25.302	38.153	27.029	33.691	0.133	25.486	38.409	27.241	33.923	0.174
+ FsCIT	**27.946**	**41.782**	**30.027**	**37.175**	**0.257**	**28.487**	**42.635**	**30.455**	**37.822**	**0.481**
FSCE[25]	26.076	41.403	27.962	31.662	9.321	28.267	44.939	30.19	33.709	11.944
+ FsCIT	**29.213**	**44.32**	**32.033**	**35.651**	**9.899**	**31.49**	**48.199**	**33.969**	**37.63**	**13.068**
DeFRCN[26]	27.984	44.343	29.273	34.482	8.489	30.489	48.285	31.866	37.112	10.62
+ FsCIT	**33.125**	**51.136**	**35.179**	**40.364**	**11.407**	**35.033**	**54.11**	**37.085**	**42.398**	**12.939**



Fig. 5. Examples of detection results on the Novel split 1 of DIOR dataset using DeFRCN with FsCIT.

TABLE VII

DETECTION RESULTS (%) OF FSOD METHODS ON AI-TOD. SHOT NUMBERS ARE 10 AND 20 ON NOVEL SPLIT 1

10 Shot	Overall (8)							Base (5)			Novel (3)		
Method	AP	AP50	AP75	APvt	APt	APs	APm	AP	AP50	AP75	AP	AP50	AP75
TFA[21]	9.19	20.623	7.014	4.941	10.069	10.069	13.8	14.704	32.996	11.222	0	0	0
+ FsCIT	**10.454**	**21.229**	**8.72**	**5.583**	**11.314**	**11.7**	**17.4**	**16.726**	**33.964**	**13.951**	**0.002**	**0.004**	**0.001**
GFSD[24]	7.369	14.622	6.571	4.471	7.841	6.758	17.874	11.79	23.395	10.514	0	0	0
+ FsCIT	**12.939**	**25.228**	**11.482**	**5.944**	**13.826**	**16.357**	**21.874**	**20.703**	**40.363**	**18.372**	**0.001**	**0.002**	0
FSCE[25]	5.602	14.585	2.973	2.565	5.97	6.357	6.9	8.349	21.217	4.611	1.024	3.532	0.244
+ FsCIT	**6.876**	**15.04**	**5.194**	**3.11**	**7.172**	**7.386**	**10.2**	**10.388**	**22.394**	**7.932**	1.022	2.783	**0.63**
DeFRCN[26]	5.001	11.45	3.811	2.353	5.608	5.3	5.6	7.478	16.33	6.07	0.872	3.317	0.045
+ FsCIT	**6.187**	**12.527**	**5.67**	**3.082**	**6.577**	**6.3**	**9.9**	**9.057**	**17.357**	**8.764**	**1.404**	**4.477**	**0.513**
20 Shot	Overall (8)							Base (5)			Novel (3)		
Method	AP	AP50	AP75	APvt	APt	APs	APm	AP	AP50	AP75	AP	AP50	AP75
TFA[21]	9.286	20.224	7.471	5.015	10.226	10.5	15.1	14.857	32.358	11.954	0	0	0
+ FsCIT	**10.788**	**21.139**	**9.357**	**5.643**	**11.737**	**12.5**	**17.7**	**17.26**	**33.821**	**14.97**	**0.001**	**0.002**	**0.001**
GFSD[24]	11.674	25.209	9.293	5.439	12.58	14.074	19.129	18.678	40.333	14.869	0	0.001	0
+ FsCIT	**13.047**	**25.369**	**11.563**	**6.003**	**13.955**	**16.355**	**21.97**	**20.875**	**40.589**	**18.501**	**0.001**	**0.003**	0
FSCE[25]	5.981	14.178	4.347	3.894	6.545	4.934	7.9	8.299	19.14	6.209	2.117	5.91	1.243
+ FsCIT	**7.213**	**15.129**	**6.111**	**4.212**	**7.834**	**6.091**	**10.8**	**10.411**	**21.031**	**9.123**	1.883	5.294	1.09
DeFRCN[26]	5.504	12.036	4.567	3.274	6.043	5.4	7.4	7.913	17.021	6.627	1.488	3.727	1.134
+ FsCIT	**6.854**	**13.871**	**6.026**	**4.925**	**7.424**	**7.5**	**10.9**	**9.98**	**19.568**	**9.131**	**1.646**	**4.376**	0.851

TABLE VIII

DETECTION RESULTS OF FSOD METHODS ON AI-TOD. NOVEL SPLIT 2

Methods	10 shots					20 shots				
	Overall (8)			Base (5)	Novel (3)	Overall (8)			Base (5)	Novel (3)
	AP	AP50	AP75	AP	AP	AP	AP50	AP75	AP	AP
TFA[21]	8.048	18.277	6.527	12.877	0	8.61	19.426	7.013	13.776	0
+ FsCIT	**9.064**	**18.983**	**7.827**	**14.503**	0	**9.323**	**19.834**	**8.268**	**14.917**	0
GFSD[24]	8.419	16.675	7.355	13.47	0	8.506	16.73	7.345	13.61	0
+ FsCIT	**9.375**	**16.92**	**8.427**	**15**	0	**9.466**	**16.841**	**8.593**	**15.145**	0
FSCE[25]	5.459	13.524	3.624	7.731	1.672	5.587	14.891	4.763	7.989	1.584
+ FsCIT	**5.874**	12.531	**4.828**	**8.721**	1.128	**6.381**	**15.149**	**5.597**	**9.181**	**1.715**
DeFRCN[26]	4.841	11.197	3.748	7.184	0.936	5.054	12.663	3.565	7.535	0.919
+ FsCIT	**6.035**	**12.665**	**5.303**	**8.955**	**1.169**	**6.796**	**15.149**	**5.597**	**9.959**	**1.523**

we make the following explanations: the contrastive proposal encoding (CPE) loss function of FSCE is mainly used to relieve the misclassification issue of novel categories. The CPE contains a proposal consistency control factor $f(u_i)$ controlled by an IoU threshold. However, according to [39], small changes in the positions between proposal and ground-truth

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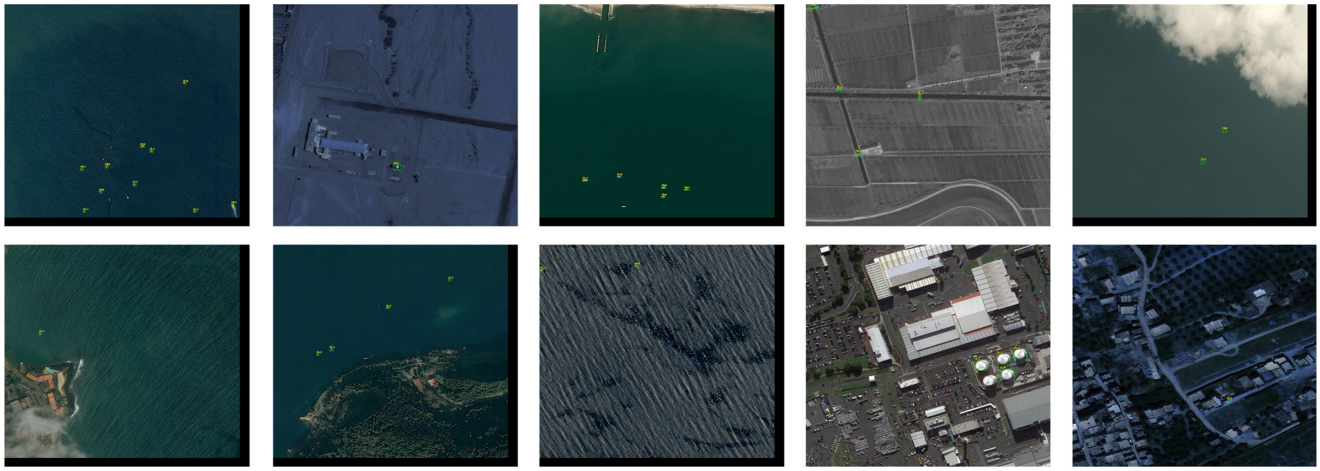


Fig. 6. Examples of detection results on the Novel split 1 of AI-TOD dataset using DeFRCN with FsCIT.

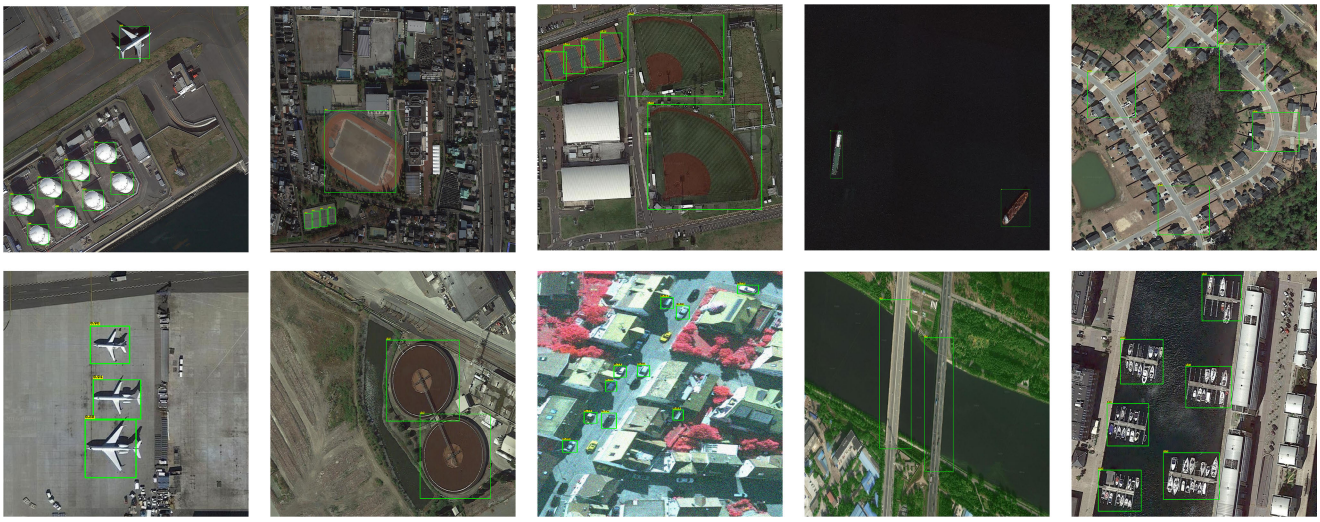


Fig. 7. Examples of detection results on the Novel split 1 of HRRSD dataset using DeFRCN with FsCIT.

TABLE IX

DETECTION RESULTS (%) OF FSOD METHODS ON HRRSD DATASET. SHOT NUMBERS ARE 10 AND 20 ON NOVEL SPLIT 1 AND NOVEL SPLIT 2

Methods	Novel split 1-10 shots					Novel split 1-20 shots				
	Overall (20)			Base (15)	Novel (5)	Overall (20)			Base (15)	Novel (5)
	AP	AP50	AP75	AP	AP	AP	AP50	AP75	AP	AP
TFA[21]	39.184	55.628	43.676	61.397	3.644	38.713	55.286	43.555	61.157	2.803
+ FsCIT	**41.192**	**58.073**	**45.500**	**63.913**	**4.837**	**41.526**	**58.448**	**45.509**	**63.893**	**5.740**
GFSD[24]	40.336	57.154	44.870	63.488	3.293	40.459	57.213	45.087	63.397	3.758
+ FsCIT	**41.815**	**58.565**	**45.765**	**64.424**	**5.641**	**42.554**	**59.652**	**46.663**	**64.457**	**7.510**
FSCE[25]	40.681	61.534	44.193	57.351	14.010	44.969	67.697	48.546	60.685	19.821
+ FsCIT	**43.369**	**65.228**	**47.152**	**59.473**	**17.602**	**45.379**	**68.701**	48.466	**61.598**	19.429
DeFRCN[26]	38.552	59.118	41.657	55.165	11.971	41.593	64.434	44.625	57.022	16.907
+ FsCIT	**41.711**	**63.505**	**44.108**	**59.142**	**13.819**	**44.408**	**67.062**	**48.209**	**61.146**	**17.628**
Methods	Novel split 2-10 shots					Novel split 2-20 shots				
	Overall (20)			Base (15)	Novel (5)	Overall (20)			Base (15)	Novel (5)
	AP	AP50	AP75	AP	AP	AP	AP50	AP75	AP	AP
TFA[21]	39.467	58.002	44.490	58.625	8.815	40.255	58.498	46.036	58.868	10.473
+ FsCIT	**43.047**	**62.300**	**49.235**	**60.125**	**15.722**	**43.363**	**62.181**	**49.024**	**60.635**	**15.728**
GFSD[24]	43.091	63.068	48.410	59.769	16.405	43.286	63.283	48.643	59.982	16.574
+ FsCIT	**44.846**	**64.417**	**50.380**	**61.187**	**18.700**	**44.798**	**64.766**	**50.442**	**61.013**	**18.855**
FSCE[25]	44.359	66.143	49.366	55.045	27.263	47.374	72.155	52.562	56.848	32.214
+ FsCIT	**45.131**	**67.915**	**49.827**	**55.794**	**28.069**	**48.534**	**73.034**	**53.778**	**57.834**	**33.654**
DeFRCN[26]	41.347	63.981	46.521	51.601	24.940	45.016	68.536	49.385	55.042	28.974
+ FsCIT	**46.725**	**69.543**	**51.477**	**56.842**	**30.538**	**48.350**	**72.113**	**53.932**	**57.869**	**33.121**

bounding box may lead to big change of IoU value of tiny objects, so $f(u_i)$ is unfriendly to tiny objects, which may be the reason for the degradation of the detection performance of tiny objects in the novel category. We visualize the detection results in Fig. 6.

HRRSD: We evaluate 10/20-shot setups on the Novel split 1 and Novel split 2 of HRRSD and report the detection performance in Table IX. The results demonstrate that our FsCIT consistently improves the baselines under two different category splits settings. Concretely, for Novel split 1 and

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ten shot, with FsCIT, DeFRCN achieves 41.711%, 59.142%, and 13.819% on AP of overall, AP of base, and AP of novel, respectively, which significantly outperforms DeFRCN by 3.159%, 3.977%, and 1.848% on AP of overall, AP of base, and AP of novel, respectively. In other data settings, the performance improvements of the remaining baselines are roughly the same. We also visualize the detection results in Fig. 7.

V. Conclusion

In this work, we propose a few-shot aerial image object detection with FsCIT to avoid filtering out proposals belonging to the novel categories in the RPN stage and ease the difficulty of tiny object detection in the aerial image. In the proposed confidence-iou collaborative proposal filtration scheme, we combine the unbiased IoU scores with the foreground–background confidence scores to filter out to filter redundant region proposals. Then, we design a new tiny object loss constraint term, which considers the central point distance, the size of ground-truth bounding boxes, and the distances between the four edges of the ground-truth bounding box and the predicted bounding box. Finally, experimental results show that our FsCIT significantly improves the prevailing FSOD methods across DIOR, AI-TOD, and HRRSD datasets. In our future work, we intend to design a few-shot oriented object detection method for more accurate detection of objects in aerial images.

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